MDM2 Group Project 1

How do varying environmental conditions affect plant-pollinator relationships in Great Britain?

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Abstract

This study aims to increase the understanding of the robustness of plant–pollinator relationships in Great Britain under changing environmental conditions to provide actionable insights for government policy. Machine-learning methods (Random Forest, XGBoost, and Linear Regression) are applied alongside a SHAP explanatory model to derive a Per Feature Importance Score (PFIS). Using 2040 climate predictions from the UK Climate Projection 2018 (UKCP18), both optimal conditions for pollination and the robustness of plant–pollinator networks are found. In a warming scenario 2 ° C, a derived robustness metric indicates severe degradation, with 46% of interactions with the three most popular UK plants collapsing and 63% substantially weakened. Temperature is the main driver, followed by the level of sunshine. Floral diversity reduces to 8%. The interactions peak at 17–22 °C with light winds and intermediate sunshine. Climate driven reductions in Rubus fruticosus (blackberry) production alone lead to a potential loss of up to 5.7 % approximately in British agriculture sales. The government should prioritize policies that increase floral diversity in landscapes and create thermally buffered habitats to strengthen networks against climate stress. In parallel, a targeted mitigation of temperature-driven declines is essential, as warming is the main driver of interaction collapse across plant-pollinator networks in the UK.

1 Introduction

Over 87% of flowering plants depend on animal pollination for reproduction (Ollerton, Winfree, and Tarrant 2011), creating a global industry worth over $3 trillion. The collapse of plant-pollinator relationships would present a catastrophic scenario for our global ecosystem. This concern is heightened by strong empirical evidence that ecosystems are already in rapid retreat as a result of human-driven climate change (Halsch et al. 2021). Gaining insight into the effects of climate features on plant-pollinator relationships and wider network robustness is a matter of national and international urgency, enabling data-driven decisions to combat the effects of a changing climate.

Plant-pollinator relations are dictated by multiple drivers. Research shows that phenology and weather heavily shape activity (Byers 2017b). Relations have evolved to their surroundings (Redhead, Woodcock, et al. 2018), leading to dependencies developing between particular plants and pollinators. As a result, one species could be adversely affected by a detriment in the other (Stang, Klinkhamer, and van der Meijden 2009). Consequently, research is focused around plant and pollinator types, weather, and phenology using climate predictions from the Office for National Statistics UK Climate Predictions (UKCP18) data from 2018 (Office 2025).

The aim of this study is to increase our understanding of the robustness of plant–pollinator relationships in a changing climate, define the optimal conditions for these relationships, and provide actionable insights to inform government policy. This will ensure resilient pollination networks that are vital to biodiversity, food security and the agricultural economy. This study differs from traditional biological analyses by applying machine learning methods to large, multi-feature datasets. This approach enables the detection of complex, non-linear relationships often overlooked by conventional methods.

2 Methodologies

This study employs a combination of Random Forest regression, SHAP value decomposition, and a bespoke climate-sensitivity metric to quantify the robustness of plant–pollinator relationships. This section outlines how the study capitalizes on recent advances in data science by using modern machine learning techniques to maximize the reliability of the results. Each model is broken down and explained in detail. More information on the implementation of these models and the code used for each can be found in the appendix section.

2.1 Assumptions

1. **Consistent data collection methods** - Equating equal entries enables ML methods to spot patterns in data.
2. 85% **similarity threshold** - Using the Rapid-Fuzz package allows for the grouping of entries with a 15% uncertainty.
3. **Observing does not affect pollinator behaviour** - Allows the study to use observations to describe pollinator behaviour.

4. **Missing data are random** - Treating missing values in the key columns (Section 2.1) as occurring at random justifies their removal without introducing systematic bias in the ecological relationships under investigation.
5. **Pollinators are not affected by climate features** - Assuming that climate change affects the locations and flowering periods of plants while pollinator traits remain constant. This simplifies the of network change under climate stressors, relying on data in (Haq et al. 2023) that pollinators show divergent reactions to climate features. The study assumes that these effects cancel.
6. **Limited climate impact during 2010–2024** - Assuming that no considerable effects of climate change have been observed between 2010 and 2024, allows the empirical network to be treated as a baseline for "current" conditions against which future scenarios are compared. This is supported by climate data, that shows no significant change is weather features during this time (Met Office 2025).
7. **Seasonality is not an external stressor** - Treating seasonality as a key driver for optimizing species interactions but not as an external stressor implies that robustness is evaluated mainly against exogenous climatic variation rather than intrinsic seasonal cycles.
8. **Non-migratory pollinators** - Assuming that pollinators do not migrate removes the need to model large-scale movement and focuses the analysis on local changes in interaction structure.
9. **Blackberry harvests scales with interaction frequency** - Assuming that commercial blackberry production is directly proportional to interaction frequency provides a simple functional link between network-level changes and an economically interpretable outcome.
10. **SHAP values as faithful explanations** - Assuming that SHAP values accurately depict the impact of each environmental feature on plants and pollinators justifies using them to interpret model behaviour and inform targeted management measures.

2.2 Dataset/Data Cleaning

The chosen dataset described plant–pollinator interactions across the UK between 2008-2024. The key columns identified for cleaning and analysis were plant and pollinator names, temporal data (week, year), environmental conditions (sunshine, wind speed, temperature), and location.

Cleaning involved removing rows with blank entries for these key features and converting qualitative data into numerical scales. This meant categorising sunshine levels from 1-5 (Appendix Section A) and converting wind speed descriptions to align with existing Beaufort-rated wind speed data shown (Appendix Section B).

Plant names were then standardised using a Python script. Re and rapidfuzz libraries were used to automate name standardisation using a list containing all Latin names for wildflowers in the UK (UK Wild Flowers 2024). The main function of the code:

1. Extracted plant names from original data set and Latin name list, removing all punctuation and converting text to lowercase.
2. Selected closest match using a similarity score; names scoring above 85%, the names were standardised with the Latin name it matched with.
3. If no match met the threshold, the entry was marked as N/A and then filtered out.

2.3 Random Forest Regression Model

A Random Forest Regressor is used to model pollination interactions as a function of different features. Random Forests (RF) minimises bias by taking the average result of numerous decision trees, which are pipelines of feature combinations. Every consideration leads to a partition using some feature that results in a more homogenous subset of interaction counts. Each partition aims to minimise the Mean Squared Error (MSE) of the interaction counts within the resulting child nodes. Through identifying correlations between features and resulting MSE, the RF is able to predict the interaction volume using feature values, and then quantify the importance of each feature in the parent node using Equation 1.

$$\text{Importance}(S, f) = \frac{N_S}{N_{\text{total}}}\,\text{MSE}(S) - \frac{N_L}{N_{\text{total}}}\,\text{MSE}(L) - \frac{N_R}{N_{\text{total}}}\,\text{MSE}(R) \tag{1}$$

Child nodes L and R are split from their parent node S on some feature, f (e.g. $f < x$ and $f \geq x$). MSE of the L and R are subtracted from the MSE of S. All MSE values are standardised by dividing their size by the combined size of all nodes in the tree N_{total}, leaving a clearly interpretable value for the effect (importance) of a feature on the MSE of S.

Using the RF model was ratified using R^2 and mean absolute error (MAE). The 18 highest interaction observations were extracted from the data. The mean interaction count, $\bar{\mu}$ was computed, and each observation, *obs*, classified as $obs > \bar{\mu}$ or $obs < \bar{\mu}$. The RF was trained on the new data using a 0.2 test-split. This provided a focused test of whether the model can reliably detect climate-driven differences in interaction strength.

2.4 SHAP analysis

Maximising the opportunities of modern ML algorithms means applying the SHAP model analysis to the RF importances list. Every RF importance value is passed through Equation 2 to return a SHAP value. ϕi represents the contribution of feature i to the RF prediction of pollination interactions. What distinguishes SHAP value from the RF importance is the consideration of the impact of other features on the impact of i. For instance, the benefit of sunshine for a plant depends on adequate rainfall to hydrate it. A SHAP value is defined by

$$\phi_i(f, x) = \sum_{S \subseteq N \setminus i} \frac{|S|!\,(|N| - |S| - 1)!}{|N|!}\left[f\left(S \cup i\right) - f\left(S\right)\right] \tag{2}$$

To aid explanation α and β are defined as

$$\alpha = \frac{|S|!\,(|N| - |S| - 1)!}{|N|!}, \qquad \beta = f\left(S \cup i\right) - f\left(S\right)$$

N is the set of all features, and S is a subset of N that excludes environmental feature, i. α weights each value relative the probability of its specific subset S, whilst β provides the effect of introducing feature i to the subset on the RF prediction. Summing every possible S gives a heavily scrutinised and validated importance value for wind-speed, temperature or sunshine, denoted ϕ_w, ϕ_t, ϕ_s, respectively.

The next step in analysing plant-pollinator relationships is to identify specific environmental and temporal conditions that maximise the number of interactions between plants and pollinators. Interactions were defined as the number of pollinators (queens, workers, males, unknown) recorded on a plant in each observation.

2.5 Network Optimisation

Feature optimisation that maximised interaction counts combined two components 1) quantitatively measure feature importances through Random Forest (RF) and a Shap model, and 2) analysing specific species-level feature patterns (temperature, seasonality). Random Forest importance values show how frequently a specific feature creates effective splits in the model's decision tree to improve the model's predictions. SHAP values describe the average contribution of magnitude to an individual prediction. As a result, SHAP values are larger for more dominant features (e.g. temperature for pollinators), while distributing the less important features more evenly. Therefore, RF importances describe how often and how impactful a feature is used in the model, while SHAP values show the feature's influence on each prediction.

2.6 Climate-Sensitivity Metric

To evaluate how external stresses will affect each plant–pollinator pair, a metric is developed that incorporates three components relevant to the robustness of plant-pollinator relations:

1. Current interaction count, given by the plant–pollinator pair's mean observed pollination interaction count
2. Projected change in each environmental driver in accordance with UKCP18 over the next 20 years
3. Magnitude and direction provided by SHAP analysis, building on RF importance scores.

Because SHAP gives the expected change in pollination interactions as a result of a one-unit change in a feature, for feature i, the Per Feature Importance Score (PFIS) is found using Equation 3 below.

$$\text{PFIS}_i = \frac{\Delta C_i \bar{\phi}_i}{\text{Interaction Count}} \tag{3}$$

where ΔC_i is feature's projected climate change, and $\bar{\phi}_i$ is the mean SHAP value for that feature for the given plant–pollinator pair over time. Using UKCP18 data (McSweeney 2015)

$$\Delta C_{\text{wind}} = 0, \qquad \Delta C_{\text{temp}} = 2, \qquad \Delta C_{\text{sun}} = 0.1$$

Since wind is not predicted to be affected by climate change over the next 20 years, it is removed as an environmental feature.

Multiplying the predicted change by the calculated values for ϕ will give us the expected population change as a result of each feature. Dividing by observed pollination interactions normalises the metric, leaving an easily interpretable representation of PFIS. PFIS values are interpreted in line with IUCN framework (IUCN 2012) as

- $-1 \leq \text{PFIS} < -0.8$: Collapse
- $-0.8 \leq \text{PFIS} < -0.3$: Reduction
- $-0.3 \leq \text{PFIS} < 0.3$: No reasonable impact
- $0.3 \leq \text{PFIS}$: Strong resulting growth

PFIS values are a decimal value of the current interaction size. Any feature contribution producing PFIS ≤ -0.8 (that is -80%) is removed to simulate the effective collapse of that plant-pollinator relationship. If all values for PFIS are greater than -0.8, then each PFIS value is summed to give a final vulnerability score, PFIS_T, that accounts for positive and negative climate impacts. PFIS_T is inversely proportional to relationship robustness, R, which is given by the inverse of the summation of Equation 4 below.

$$R = \sum_{i \in F} \left| \frac{\Delta C_i \, \bar{\phi}_i}{\text{Interaction Count}} \right|^{-1} \tag{4}$$

This framework provides a quantitative basis for identifying vulnerable relationships and guiding conservation interventions aimed at supporting flora and fauna under future climate scenarios.

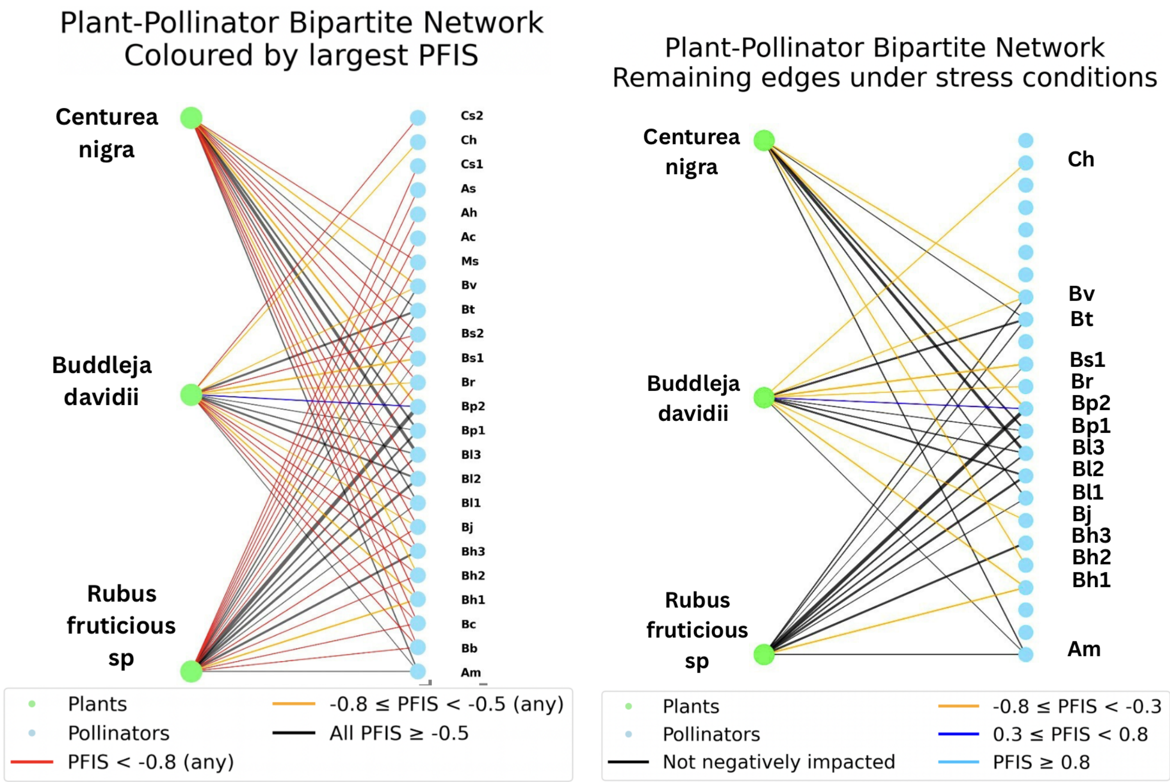
3 Results

Applying the methods of the methodologies section to empirical data enables the model to make predictions for the real world. Section 3.1 will investigate the effects of changing climate features of current plant-pollinator networks. Section 3.2 will explore the importance of different features to support advice for government, public trusts, and private corporations regarding pollination networks.

3.1 Robustness

Figure 1a shows a bipartite graph of the plant-pollinator network for the top three plants, with edge thicknesses proportional to the robustness value (R) of each edge. Figure 1b presents the same 3 plant network under the projected environmental conditions. Across the 3 plant network, 48% of plant-pollinator interactions fall below the collapse threshold ($R < -0.8$), and 57% exhibit reductions outside the $\pm 50\%$ natural fluctuation range.

Under temperature changes, 17 edges fall below the collapse threshold, while sunshine changes remove 16 edges. 5 edges meet this threshold as a result of both drivers. Sunshine reduces interaction instances by $30\% - 80\%$ in 12 edges, and temperature produces 5. Pollinator label translations are found in the appendix, Section C.



(a) Collapsed and reduced edges

(b) Unaffected and reduced edges

Figure 1: Effect of UKCP18 on pollination network of top 3 plants by interaction count: (a) shows collapse or reduced edges; (b) shows unaffected and reduced edges.

Extending the model to the top 10 plants results in 177 interactions, accounting for 30% of the empirical data. 27 interactions fall below the collapse threshold. Temperature accounts for 19 of these removals, and sunshine accounts for 8. Overall, 27% of edges are negatively affected, while less than 2% are positively affected. Figure 3 visualises the resulting network. For datasets containing the 30 and 50 plants with the highest degrees, temperature removals tend towards 5% of edges and 1% for sunshine removals. This trend is seen in Figure 2.

Sensitivity analysis tested these results by perturbing the UKCP18 projected climate deltas by $\pm 5\%$ and $\pm 10\%$. The negative impact graphs for each result are given in Appendix G. In each case temperature remained the dominant collapse driver, while the number of collapsed edges varied by less than 2.5%, confirming that the results are robust to small changes in ΔC.

In addition to perturbing climate forecasts, the RF model used in calculating PFIS was evaluated on the top 5 highest-interaction observations for the most frequently recorded plant–pollinator pair to reduce noise. The RF achieved an R^2 of 0.60 and a mean absolute error (MAE) of 0.29 in predicting above or below mean interaction counts. Together with the sensitivity analysis above, this quantitative validation shows that both the robustness metric and its underlying RF model capture meaningful, climate-driven variation in interaction strength.

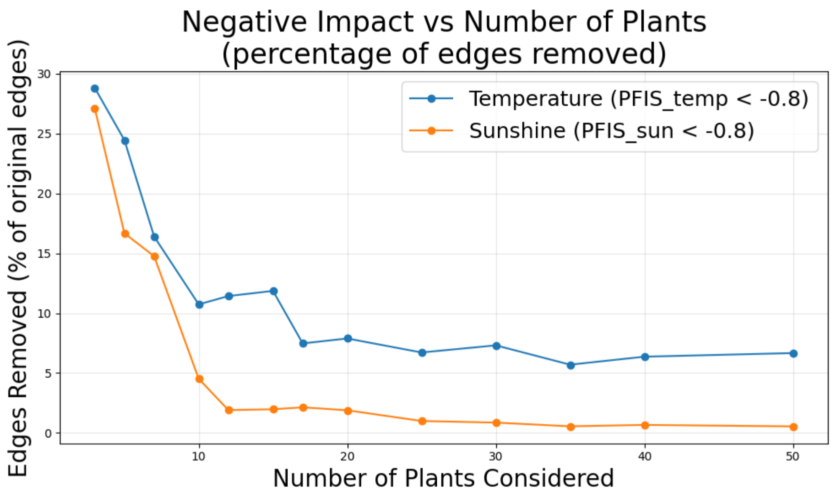


Figure 2: Collapsed and reduced edges.

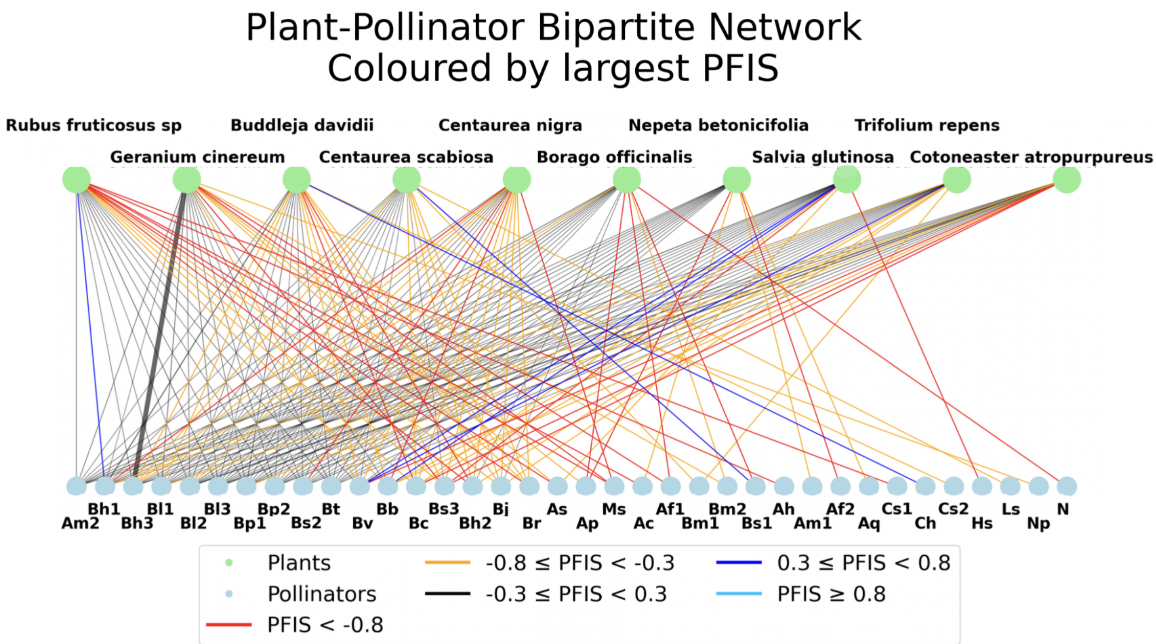


Figure 3: Effect of UKCP18 on pollination network of top 10 plants by interaction count

3.2 Optimisation

To provide climate conditions that governments should aim to foster and protect, this section examines ideal conditions UK for flora and fauna networks. Table 1 exhibits the mean values for the key environmental and temporal feature importances of the top three plants and pollinators obtained through two models.

Table 1: Comparison of feature influences for Top Three Plants and Pollinators for both RF and SHAP mean importances.

	Top 3 Plants		**Top 3 Pollinators**	
Feature	**Random Forest**	**SHAP**	**Random Forest**	**SHAP**
Temperature	0.2841	79.03	0.7881	397.62
Wind Speed	0.2099	36.65	0.0666	34.12
Sunshine Level	0.1544	30.11	0.0430	21.10
Seasonality (Sin + Cos)	0.3516	103.67	0.1023	42.40

Looking at the RF importances, temperature was a driving feature for pollinator interactions, with its RF importance value (0.7881) being nearly eight times greater than seasonality (0.1023). Seasonality had a stronger impact on plants compared to pollinators, with an RF importance of 0.3516. Looking at the normalised SHAP values within each group, temperature contributed $\approx 80\%$ of total SHAP importance for pollinators, displaying weak sensitivity to seasonality, contributing $\approx 9\%$. This disparity was not replicated with the normalised SHAP importances for plants. Plants distributed a more balanced distribution of feature importances. Temperature contributed $\approx 32\%$ with seasonality being the driving feature, contributing $\approx 42\%$. Wind speed and sunshine accounted for $\approx 27\%$ of total feature importance for plants while only accounting for $\approx 11\%$ for pollinators.

The top three plants and pollinators were analysed to identify peak environmental and temporal ranges that would result in the optimal number of interactions. Values for the optimal conditions can be seen in Table 2 for top three plants, and Table 3 for top three pollinators.

Table 2: Optimisation Summary for Top Plants

Attribute	**Rubus Fruticosus**	**Centaurea Nigra**	**Buddleja Davidii**
Group	Plant	Plant	Plant
Total Interactions	2536	1448	1402
Optimal Weeks	22–26	30–34	27–32
Optimal Temperature (°C)	17–22	18–22	18–22
Optimal Wind (Bf)	2–3	2–3	1–3
Optimal Sunshine	3	3	3
Dominant Driver	Temperature $\approx$ Seasonality	Temperature	Seasonality

Table 3: Optimisation Summary for Top Pollinators

Attribute	**Bombus Pascuorum**	**Bombus Terrestris**	**Apis Mellifera**
Group	Pollinator	Pollinator	Pollinator
Total Interactions	10223	6316	5005
Optimal Weeks	32–37	28–34	22–39
Optimal Temperature (°C)	17–22	16–22	18–23
Optimal Wind (Bf)	1–3	2–3	2–3
Optimal Sunshine	3	3	3
Dominant Driver	Temperature	Temperature	Temperature

Tables 2 and 3 summarise the optimal conditions for the top three plants and pollinators. Temperature and seasonality were equally influential in determining total plant interactions, while temperature was the only dominant driving feature. The top plant species exhibited an optimal number of interactions between 17°C –22°C, while pollinators showed similar but broader optimal temperature ranges. Majority of both plant and pollinator species preferred moderate wind conditions (Beaufort 1-3). Across all species, sunshine level 3 was associated with the highest interaction count. Optimal weeks differed slightly between species, with pollinator peak activity lagging behind plant peak activity.

In Figures 4 and 5 species interactions are plotted against different temperatures which show the optimal temperature ranges for interactions for *Rubus fruticosus* and *Bombus pascuorum.*

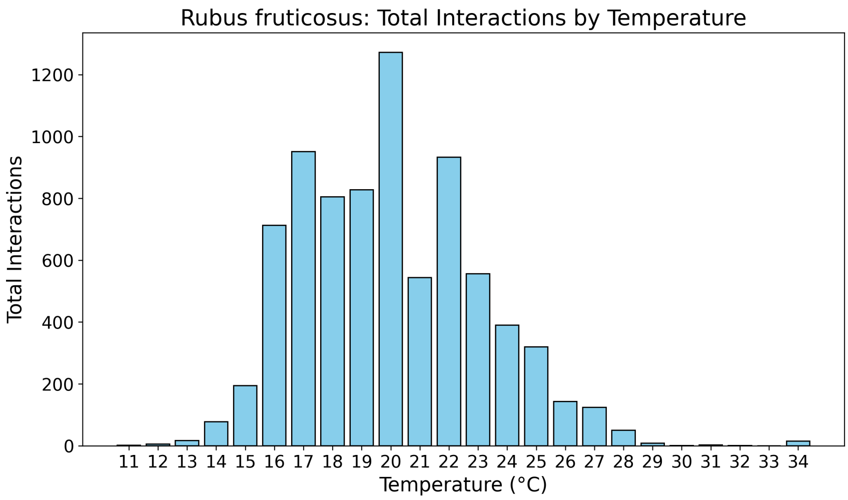


Figure 4: Temperature sensitivity (Temperature vs Total Interactions) for *Rubus Fruticosus.*

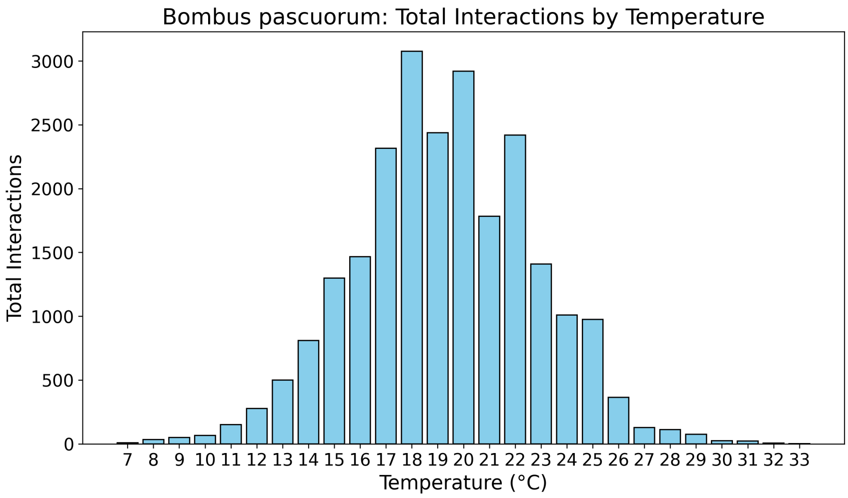


Figure 5: Temperature sensitivity (Temperature vs Total Interactions) for *Rubus Pascuorum*.

Interaction numbers for *Rubus fruticosus* increase from 15°C onwards with the most interactions occurring between 17–22°C, accounting for around 60% of all interactions. The interaction peaks at 20°C which accounted for 14.2% and then steadily declined after 23°C. Interactions began increasing steadily at around 11-12°C, then rising sharply between 14-17°C. The majority of interactions occurred between 17-22°C, representing approximately 62%. The number of interactions peaked at 18°C which represented 12.93%, and interaction numbers began decreasing steadily from 21°C onwards.

Figures 6– 7 display top plant and pollinator interactions at each week throughout the year, exhibiting key flowering and foraging periods for plants and pollinators accordingly.

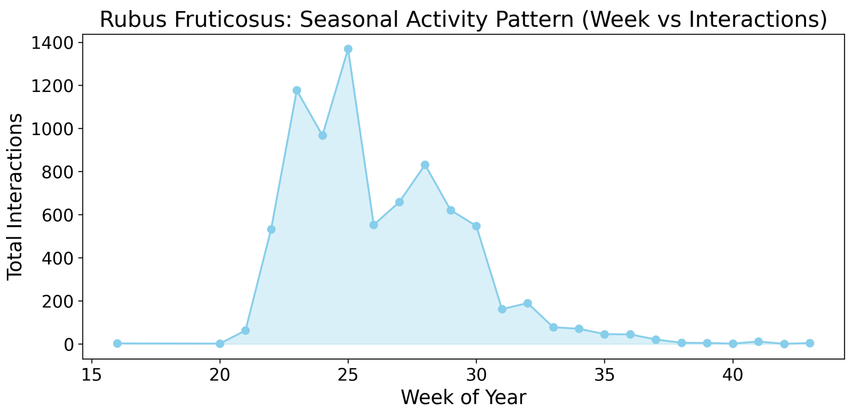


Figure 6: Seasonal activity pattern (Week vs Total Interactions) for *Rubus fruticosus*.

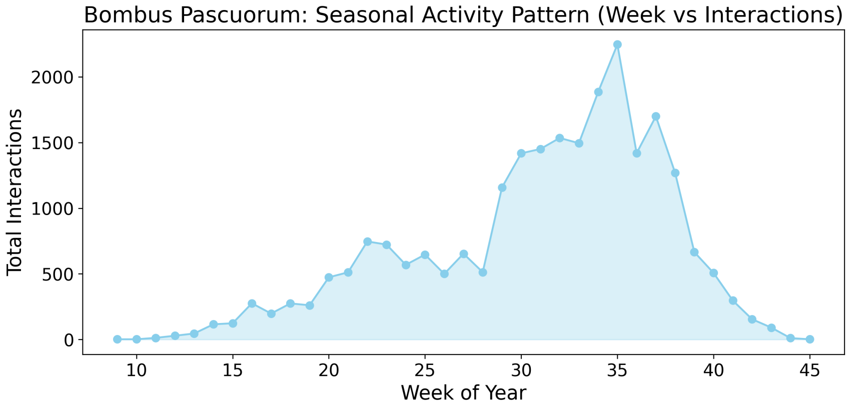


Figure 7: Seasonal activity pattern (Week vs Total Interactions) for *Bombus pascuorum*.

In Figure 6, *Rubus fruticosus* interactions begin to increase around Weeks 20. Peak plant interactions took place between Weeks 22-26, with activity sharply declining by Week 32. In Figure 7, *Bombus pascuorum* begin to increase from Week 14 onwards with peak foraging periods occurring between Weeks 32-37. Activity substantially diminishes by Week 42.

4 Discussion

Comparing results with existing studies builds an understanding of the model's accuracy and data interpretation aids in providing accurate insights for policymakers.

4.1 Robustness

The robustness results provide an in depth evaluation of how the UK plant-pollinator network reacts to projected climatic shifts, directly addressing this study's aim of understanding the effect of varying environmental conditions on plant-pollinator relationships in Great Britain. Using the PFIS metric defined in Equation 3 and the robustness measure in Equation 4, the application of UKCP18 projections result in major declines in interactions across the network.

Figures 1a and 1b show that under temperature changes alone, 17 edges collapse compared to 16 sunshine-driven collapses, with 5 collapsed by both. Inversely, sunshine reduces pollination instances by $30\% - 80\%$ in 17 edges, while temperature causes this reduction in only 5 edges. These patterns establish the foundation for interpreting how the network's structural stability is changed under climate pressure.

Losses are centred around high-degree plant species. Figure 2 shows an exponential decay in interaction counts as lower-degree plants are added to the model. Collapses around structurally important plants are consistent with past studies (Redhead, Dreier, et al. 2018). These plants support disproportionately more pollinator species, so interaction reductions trigger cascading declines across the entire network, an effect seen in Figure 3.

Examining larger networks reveals that flora diversity mitigates, but does not eliminate, climate-driven decline. In networks containing the top 30 and 50 plants, temperature driven collapses tend towards an asymptote of 5% and sunshine 1%, as seen in Figure 2. This reflects the stabilising effect of varied species, consistent with ecological network theory (Olesen et al. 2007).

4.2 Optimisation

Looking at the RF values, 78.81% of all useful splits in the random forest model for pollinator interactions were for temperature. This is reflected in the corresponding SHAP value of 397.62 ($\approx$ 80% of total pollinator SHAP importance). This SHAP value is extremely high, partly due to pollinators having more interactions and because it reflects pollinator sensitivity to temperature changes. Seasonality informs when pollinators could be active, while temperature indicates whether or not pollinators will be active (King, Nicolson, and Garibaldi 2022a).

The dominant drivers for plants were seasonality (RF: 0.3516, SHAP: $\approx$ 42% of total SHAP importance) and temperature (RF: 0.2841, SHAP: $\approx$ 32%), being a secondary driver. Whilst pollinator hibernation depend on temperature (King, Nicolson, and Garibaldi 2022b), plants rely primarily on the length of the day to regulate flowering (Osnato et al. 2022). This explains why temperature dominates collapses, disproportionately affecting pollinator activity relative to plant flowering cycles, causing the observed 10-week lag between seasonal peaks in Figures 7 & 6. Such a lag is consistent with bumblebee colony cycles (Bumblebee Conservation Trust 2025b). These differences drive pollination deficits (**Gerard2020**), (Byers 2017a), clearly supported by the observed difference in Figure 2.

Despite the difference in seasonality and temperature importance for plants, feature influences in terms of RF and % of total SHAP importances are more balanced. The RF values show that wind speed and sunshine are used four times more often in the plant interaction predictions compared to pollinators. Temperature overshadows all the other features for pollinators, while secondary features, including wind speed and sunshine, still play a significant role in determining plant interaction numbers.

Both species thrive in wind speed at Beaufort scale 2, which is approximately 4-7 mph (Appendix B), sunshine levels of 3 and similar temperature ranges of 17-23°C. According to (Walter et al. 2013), plants from the *Rubus* genus achieve close to ideal performance activity between the ranges 15-20°C, which aligns with the plotted behaviour for total interactions by behaviour in Figure 4. According to Springer, species within the *Bombus* genus, foraging activity increases rapidly from around 10°C (Dol and Huvermann 2020) with activity beginning to drop off at around 27°C (Bumblebee Conservation Trust 2025a). These findings match the patterns observed in the plot for *Bombus Pascuorum* shown in Figure 5.

4.3 Real-World Impacts

Climate change is expected to increase plant – pollinator temporal misalignments, reducing production, as many plant species follow suitable climates northward, while their pollinators may respond more slowly or remain relatively stationary (Haq et al. 2023).

Focusing on *R. fruticosus*, as it has the highest volume of pollination interaction and the longest flowering period among plants, any bioclimatic changes will likely propagate through the pollination network and have an impact on production (Trivedi et al. 2008). *R. fruticosus* produces blackberries, which have a well-established commercial pathway in the UK fresh-fruit and processing supply chains. implying the relationship "interactions → production". For *R. fruticosus*, temperature emerges as the most influential predictor of pollination interaction.

Figure 8 predicts pollination interactions using the XGBoost model, where the orange line is the projection by data from 2012 to 2024, which implies a gradual decline around 1000; the blue line is the prediction by using data from 2019 to 2024, which demonstrates relatively high and stable interaction levels around 1400 for the next three years, followed by a sharp decrease and subsequent stabilization at a lower level around 1200 within five years.

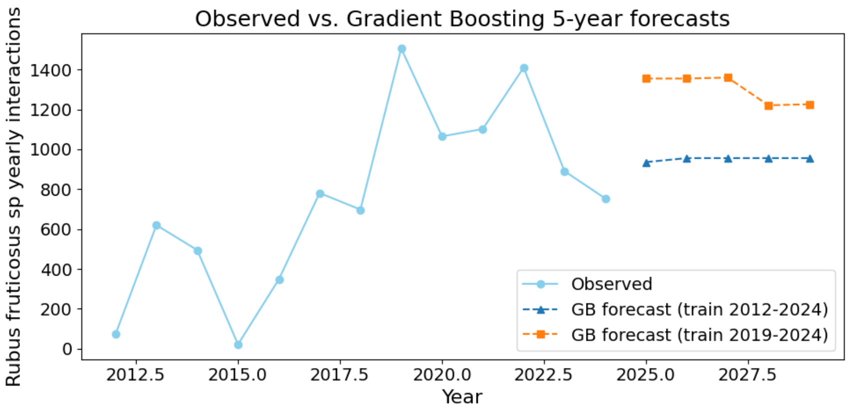


Figure 8: XGBoost Interaction Amount Prediction

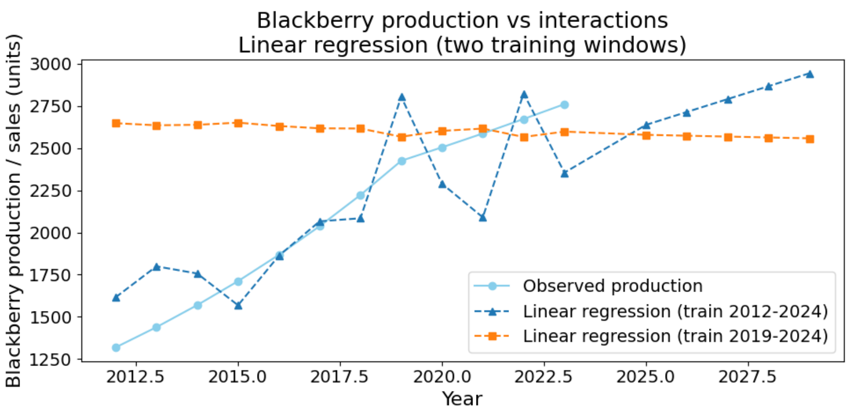


Figure 9: Linear-Regression Black Berry Harvest Amount Prediction

Figure 9 shows the prediction of production using linear regression, where the blue line is the prediction based on 2012 to 2024 data, indicating that production will increase to the highest point (2750 tonnes). The orange line is the prediction based on 2019 to 2024 data, implying production will decrease from approximately 2700 to 2500 tonnes.

Quantifying the dependence of wild plant pollination on key species such as R. fruticosus enables a more targeted management of farm margins (Klein et al. 2007). Temperature-driven projections of plant–pollinator interactions further allow the timing and spatial arrangement of these resources to be adjusted under future climate scenarios to deal with phenological mismatches and climate-driven range changes (Gérard et al. 2020). These measures can increase the resilience and efficiency of pollination services.

4.4 Limitations and Improvements

The first limitation is the dataset. Entries for sunshine were descriptive, introducing recorder subjectivity. The data is also concentrated in certain regions, but the effects of this are limited, as the UK climate is uniformly moderate outside of the north - south extremes. Descriptive entries are a bigger challenge. Future studies must use objective weather data aid pattern recognition and correct spatial bias.

Removing unrecognised plants from the data limits the realism of the model, introducing bias into the dataset unless removals were purely random. Plant name standardisation may have misclassified species due to the 85% threshold. This would adversely affect lesser known plants who were not in the reference data, possibly because of small numbers, limiting their significance. Incorporating algorithms like k-Nearest Neighbours could reduce removals and diversify data.

Past research supports temperature and sunshine as key climate drivers, but also suggests extreme weather events may dominate climate change effects. However, their unpredictability would add large uncertainty and complexity to the model, offsetting the benefits of including them. Future work should therefore focus on carefully modelling and researching the inclusion of extreme events.

Pollinators are assumed to be unaffected by climate change. Although pollinator responses are slower compared to plants, their phenology still changes (Haq et al. 2023). Therefore, results obtained for robustness may not fully capture real-world behaviour. Over the considered 20 year window this heavily impedes this models' legitimacy. Relaxing this assumption would allow the simulated ecosystems to be more adaptable, enabling better informed advice for environmental policy.

5 Conclusion

This study aims to assess how different environmental features influence plant–pollinator interactions in the UK and how robust the current pollination network is under projected climate change. Our model predicts that, under warming scenarios, nearly half (48%) of the interaction edges will become functionally lost and most of the remaining will be weakened beyond the $\pm 50\%$ range of natural variability, with temperature as the primary driver and other weather variables as secondary. Interactions are maximized under moderate temperatures (17–22 $°C$), light winds (Beaufort 1–3), and intermediate sunshine, but plants and pollinators exhibit offset seasonal peaks, creating a substantial risk of phenological mismatch that is particularly acute around high-degree plant species and may translate into decreased harvest production. These findings highlight the need to prioritize the conservation of key plant species and their habitats, enhance floral diversity to buffer climate-driven losses, consider northward shifts or other related adjustments in habitat management, and apply temperature-based projections to improve crop management and reduce temporal mismatches (Miyashita et al. 2023 Maeda et al. 2023), thus helping to prevent major declines in plant production and support a more climate-resilient pollination network.

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Appendix

A Quantified Sunshine Levels

Table 4: Quantified Levels of Sunshine.

Number	Description
1	Overcast
2	Cloudy
3	Sunny / Cloudy
4	Partly Sunny
5	Sunny

B Quantified Windspeed Levels

Table 5: Quantified Levels of Windspeed using Beaufort Scale (Bf).

Bf	Description	Land Condition	Windspeed (mph)
0	Calm (No wind)	Smoke rises vertically	<1
1	Light air	Slight smoke drift	1–3
2	Light breeze	Wind felt on face; leaves rustle	4–7
3	Gentle breeze	Leaves and twigs in constant motion	8–12
4	Moderate breeze	Dust raised; small branches move	13–18
5	Fresh breeze	Small trees in leaf begin to sway	19–24
6	Strong breeze	Large branches move; trees sway	25–31

C Pollinator translations for Figure 1

Table 6: Pollinator short labels and corresponding full species names.

Short Label	**Full Name**
Ac	Andrena cineraria
Ah	Andrena haemorrhoa
Am	Apis mellifera
As	Andrena sp.
Bb	Bombus bohemicus
Bc	Bombus campestris
Bh1	Bombus hortorum
Bh2	Bombus humilis
Bh3	Bombus hypnorum
Bj	Bombus jonellus
Bl1	Bombus lapidarius
Bl2	Bombus lucorum s.l.
Bl3	Bombus lucorum/terrestris
Bp1	Bombus pascuorum
Bp2	Bombus pratorum
Br	Bombus rupestris
Bs1	Bombus sp.
Bs2	Bombus sylvestris
Bt	Bombus terrestris
Bv	Bombus vestalis
Ch	Colletes hederae
Cs1	Coelioxys sp.
Cs2	Colletes sp.
Ms	Megachile sp.

D Interaction count for plants

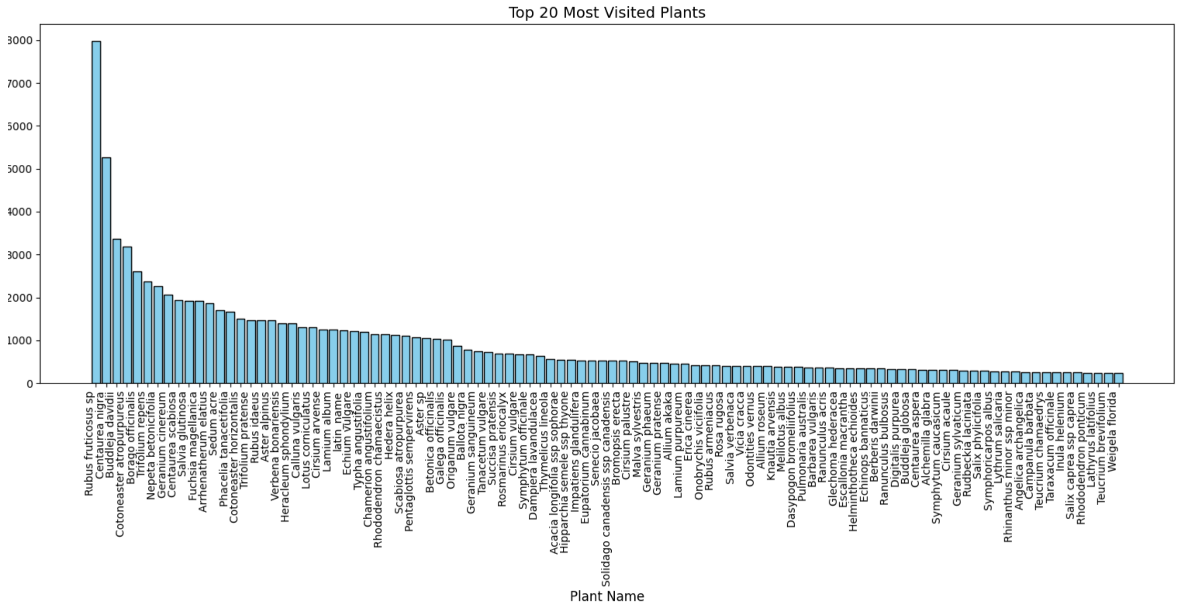


Figure 10: Figure showing decreasing returns of considering more plants

E History R. fruticosus Production for Figure 10

Year	2012	2013	2014	2015	2016	2017	2018
Production	1318	1438	1569	1711	1867	2037	2222
Year	2019	2020	2021	2022	2023	2024	
Production	2425	2505	2587	2673	2761	2779	

Table 7: History R. fruticosus(Black Berry) Harvest Amount

F Top Feature of R.Fruticosus Pollination

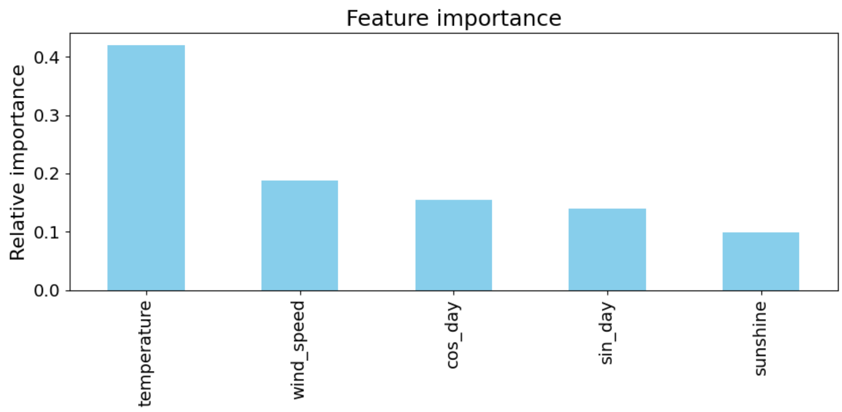
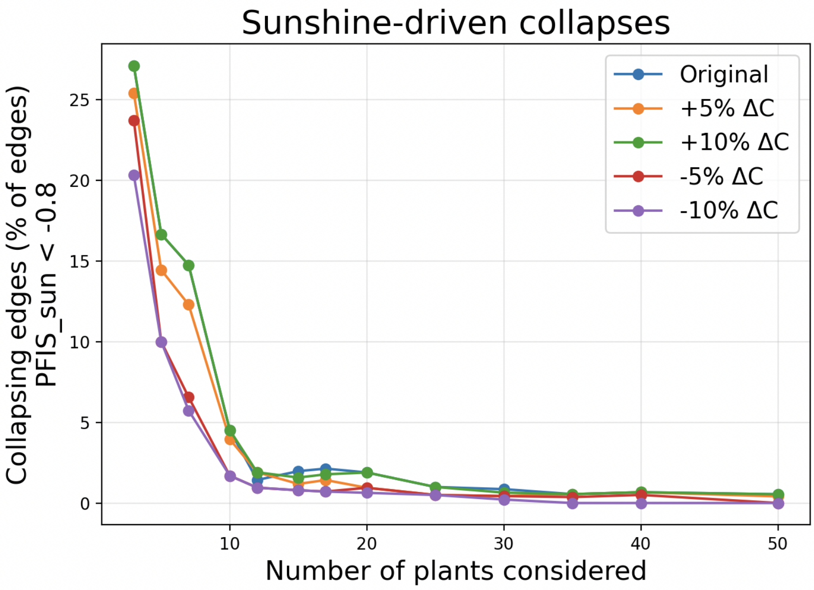
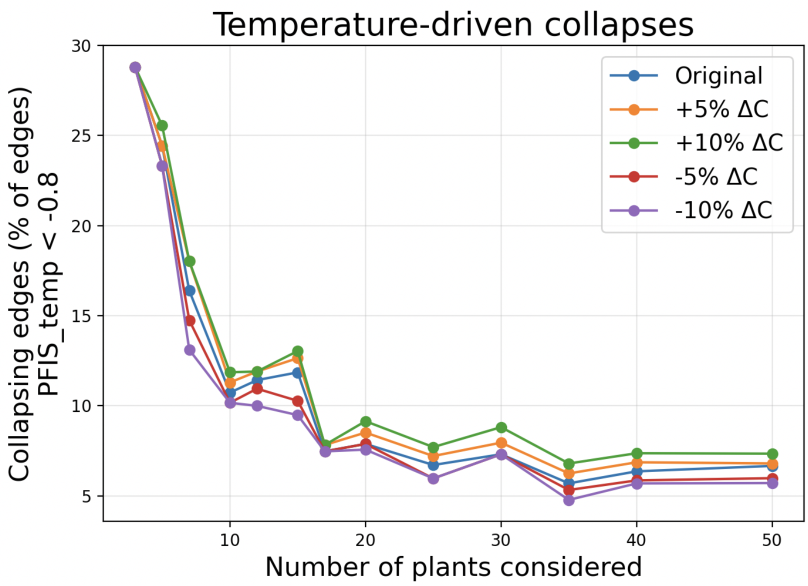


Figure 11: Top Feature of R.Fruticosus Pollination

G Effects of perturbing ΔC



(a) Figure Ga: Temperature driven collapses under varying predicted changes



(b) Figure Gb: Sunshine driven collapses under varying predicted changes